

Shangkun Jiang

Civil and Coastal Engineering, University of Florida

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Education

- **University of Florida** Aug. 2023 - Expected Dec. 2026
Ph.D. in Civil Engineering (Transportation); Minor in Statistics Gainesville, FL, USA
 - Advisor: Dr. [Xilei Zhao](#)
 - Committee members: Dr. [Lily Elefteriadou](#), Dr. [Siva Srinivasan](#), Dr. [Sayar Karmakar](#)
 - Dissertation: Advancing Transportation Systems Analysis with Emerging Mobility Data: From Operational Utility to Data Quality
- **Southeast University** Sept. 2020 - Jun. 2023
M.S.E. in Traffic and Transportation Nanjing, China
 - Advisor: Dr. [Peng Liao](#)
 - Dissertation: Research Methods on the Capacity of Inland Waterways Based on Vessel Traffic Flow Theory
- **Dalian University of Technology** Sept. 2016 - Jun. 2020
B.E. in Transportation Engineering Dalian, China
 - Advisor: Dr. Ronghan Yao

Experience

- **McTrans center** May. 2026 - Present
Research Assistant, University of Florida Transportation Institute Gainesville, FL, USA
- **University of Florida** Aug. 2023 - Present
Research Assistant, Department of Civil and Coastal Engineering Gainesville, FL, USA
- **Southeast University** Sept. 2020 - Jun. 2023
Research Assistant, School of Transportation Nanjing, China

Research Interests

Transportation: Big data, Data bias, Computational modeling, Travel behavior analysis, Urban traffic

Artificial Intelligence: Causal inference, Machine learning, Deep learning, Statistical analysis

Disaster management: Evacuation, Emergency traffic modeling, Transportation resilience

Research Projects (Selected)

- **GeoAI-Based Activity Inference from GPS Trajectories** May. 2026 - Present
University of Florida
 - Developed **GeoAI-Act**, a two-stage GeoAI framework that converts noisy GPS trajectories into semantically labeled stay episodes using Transformer-based stay detection and LLM-based purpose inference.
 - Engineered trajectory, temporal, data-quality, and OSM-based geographic features from large-scale mobility datasets to support robust stay detection and external validation across different regions and data sources.
 - Evaluated the framework against rule-based, clustering-based, and supervised baselines, achieving **0.873 point-level F1** and **0.813 segment-level F1** for stay detection, with improved stay purpose inference using **Llama-3.1 + QLoRA**.
- **Assessing spatial bias of mobile device location data** Sept. 2025 - Present
University of Florida
 - Developed a framework to evaluate spatial bias in **large-scale mobility datasets**.
 - Analyzed **10M+ anonymized mobile device** users across Florida.
 - Identified **systematic underrepresentation** in socially disadvantaged and urban areas, highlighting limitations of mobility datasets for transportation analysis.
- **Analyzing Wildfire Evacuation Behavior Using Facebook Data** Sept. 2024 - Feb. 2025
University of Florida

- Analyzed **evacuation behavior** during the 2025 Palisades and Eaton wildfires using **Facebook data**.
 - Quantified **evacuation compliance rates**, travel distances, and destination types during wildfire emergencies.
 - Identified **spatial disparities** in evacuation behavior across communities with different socioeconomic variables.
 - Demonstrated the use of **large-scale mobility data** to support disaster evacuation transportation analysis.
- **Impact of Evacuation Orders on Traffic Conditions (Hurricane Ian)** May. 2024 - Sept. 2024
University of Florida
- Evaluated **traffic impacts** of evacuation orders during Hurricane Ian using **high-resolution mobility datasets**.
 - Developed methods to estimate **space-mean speed** and **planning time index** from mobile device location data.
 - Applied **causal model** to estimate effects of evacuation orders on traffic performance.
 - Identified a 14.9% decrease in average traffic speed and a 14.6% increase in travel time variability.
- **Real-Time Urban Traffic Monitoring Using Transit Buses as Probes** Jul. 2023 - Dec. 2023
University of Florida
- Developed algorithms to estimate network-level traffic speeds using GTFS realtime transit data.
 - Designed methods for bus segment-trip extraction and speed estimation across large urban transit networks.
 - Validated results against Bluetooth sensor and Google traffic data.
- **Vessel Traffic Flow Modeling in Inland Waterways** Jan. 2023 - June. 2023
Southeast University
- Analyzed vessel traffic flow dynamics using multi-source maritime traffic datasets.
 - Investigated relationships between vessel speed, spacing, and traffic density.
 - Developed a piecewise fundamental diagram model to characterize vessel traffic flow states.

Publications

B=Book Chapter, C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

- [J] **Jiang, S.**, Lovreglio, R., Cova, T. J., Park, S., Xu, S., & Zhao, X.* (2026). *Wildfire evacuation analysis using Facebook data: Evidence from Palisades and Eaton fires*. **Transportation Research Part D: Transport and Environment**, 158, 105456. <https://doi.org/10.1016/j.trd.2026.105456>
- [J] **Jiang, S.**, Zhang, X., Wong, W.*, Park, S., & Zhao, X. (2026). *Examining the short-term impact of evacuation orders on traffic conditions: Evidence from Hurricane Ian*. **International Journal of Disaster Risk Reduction**, 143, 106288. <https://doi.org/10.1016/j.ijdrr.2026.106288>
- [J] **Jiang, S.**, Sun, Y., Wong, W.*, Xu, Y., & Zhao, X. (2025). *Real-time urban traffic monitoring using transit buses as probes*. **Transportation Research Record**, 2679(2), 219-236. <https://doi.org/10.1177/03611981241260708>
- [B] **Jiang, S.***, Zhang, X., & Zhao, X. (2026). *Review of multi-source data for disaster management*. In X. Huang, J. Shi, & M. Yang (Eds.), **Artificial Intelligence in Safety Science and Engineering** (Chap. 8, pp. 183–213). Elsevier. <https://doi.org/10.1016/B978-0-443-36342-9.00008-6>
- [J] Yang, W., **Jiang, S.**, Liao, P.*, & Wang, H. (2023). *Evaluation of car-following model for inland vessel-following behavior*. **Ocean Engineering**, 284, 115196. <https://doi.org/10.1016/j.oceaneng.2023.115196>
- [J] Liu, L.*, Zhang, X., **Jiang, S.**, & Zhao, X. (2025). *Hurricane evacuation analysis with large-scale mobile device location data during hurricane Ian*. **Transportation Research Part D: Transport and Environment**, 139, 104559. <https://doi.org/10.1016/j.trd.2024.104559>
- [J] Yang, W., Liao, P.*, **Jiang, S.**, & Wang, H. (2025). *Analysis of vessel traffic flow characteristics in inland restricted waterways using multi-source data*. **Ocean Engineering**, 317, 120065. <https://doi.org/10.1016/j.oceaneng.2024.120065>
- [J] Recalde, A.*, Liu, L., Zhang, X., Park, S., **Jiang, S.**, & Zhao, X. (2026). *Evacuation Destination Choices during Hurricane Ian: A Direct Demand Modeling Approach*. **Transportation Research Part D: Transport and Environment**, 156, 105384. <https://doi.org/10.1016/j.trd.2026.105384>
- [C] **Jiang, S.**, & Jin, C. J.* (2023, February). *Microscopic simulations of traffic congestion in Runyang Bridge: comparisons between two cases*. In Sixth **International Conference on Traffic Engineering and Transportation System (ICTETS 2022)** (Vol. 12591, pp. 865-870). SPIE. <https://doi.org/10.1117/12.2668553>
- [S] Garces, S., **Jiang, S.**, Lovreglio, R.*, Kuligowski, E., Cova, T.J., Xu, S., & Zhao, X. (2025). *A Systematic Literature Review for Wildfire Emergency Management Platforms*. Manuscript submitted for **Fire Technology**.
- [S] Liao, P.*, Yang, W., **Jiang, S.**, & Wang, H. (2024). *Evaluation of waterway lock service quality in Yangtze Delta: from the perspectives of customer and supplier*. Manuscript submitted for publication in **Maritime Policy & Management**. <https://doi.org/10.48550/arXiv.2410.07132>

PRESENTATIONS & TALKS

- [1] **Jiang, S.**, Sun, Y., Wong, W.^{*}, Xu, Y., & Zhao, X. (Jan, 2024). *Real-time Urban Traffic Monitoring By Using Transit Buses As Probes*. **Lectern** session on **Transportation Research Board 103th**, Washington, DC.
- [2] **Jiang, S.**^{*}, Lovreglio, R., Cova, T.J., Park, S., Xu, S., & Zhao, X. (Jan, 2026). *Analyzing Wildfire Evacuation Behavior Using Facebook Data: A Case Study of the 2025 Palisades and Eaton fires*. **Poster** session on **Transportation Research Board 105th**, Washington, DC.
- [3] **Jiang, S.**^{*}, Zhang, X., & Zhao, X. (Jan, 2026). *A Systematic Review of Multi-Source Data for Disaster Management: Strengths, Limitations, and Integration Strategies*. **Poster** session on **Transportation Research Board 105th**, Washington, DC.
- [4] Sun, Y.^{*}, **Jiang, S.**, Lovreglio, R., Xu, S., Cova, T.J., & Zhao, X. (Jan, 2026). *Using Survey, GPS, and Facebook Data to Analyze Wildfire Evacuation*. **Poster** session on **Transportation Research Board 105th**, Washington, DC.
- [5] Park, S., **Jiang, S.**, Sun, Y., & Zhao, X.^{*} (Jan, 2026). *Developing a Dynamic Social Vulnerability Index for Public Health (DSVI-PH) Using Location-Based Data in Hurricane Beryl*. **Poster** session on **Transportation Research Board 105th**, Washington, DC.
- [6] Yang, W.^{*}, **Jiang, S.**, Liao, P., & Wang, H. (Jan, 2025). *Assessment of Waterway Lock Service Quality: Perspectives from Customers and Suppliers*. **Poster** session on **Transportation Research Board 104th**, Washington, DC.
- [7] Bella, G.^{*}, Tree, D., **Jiang, S.**, Zhao, X., Palm, M., Miller, R., Grajdura, S. (March, 2026). *Understanding Evacuation Dynamics through Diverse Mobility Datasets: Lessons from the Eaton and Palisades Fires*. **American Association of Geographers (AAG) Annual Meeting 2026**, San Francisco, CA.
- [8] Yan, X.^{*}, Recalde, A., **Jiang, S.**, Zhang, X., Zhao, X. (March, 2026). *Understanding Evacuation Dynamics through Diverse Mobility Datasets: Lessons from the Eaton and Palisades Fires*. **American Association of Geographers (AAG) Annual Meeting 2026**, San Francisco, CA.
- [9] Liu, L.^{*}, Zhang, X., **Jiang, S.**, & Zhao, X. (Jan, 2025). *Hurricane Evacuation Analysis with Large-scale Mobile Device Location Data*. **Poster** session on **Transportation Research Board 104th**, Washington, DC.
- [10] Zheng, W.^{*}, Lee, D., Chen, X., Dong, Z., & **Jiang, S.**, (Jan, 2024). *Behavior Mechanism of Travel Mode Choice During Large-scale Events*. **Poster** session on **Transportation Research Board 103th**, Washington, DC.

Media Coverage

- [1] [From the SERMOS Lab to McTrans, Shangkun Jiang's next chapter](#). University of Florida Transportation Institute, Student Spotlight, June 8, 2026.

Skills

- **Programming Languages:** Python, R, C, C#
- **Database Systems:** MySQL, MongoDB
- **Transportation Software:** ArcGIS / QGIS, AutoCAD, HCS, HSS, VISSIM, SUMO, TransCAD
- **Mathematical & Statistical Tools:** STATA, AMOS, SPSS
- **Research Skills:** Academic Writing, Machine Learning, Deep Learning
- **Language:** English (Fluent), Mandarin (Native)

Academic Service

Reviewer for:

- [Applied Ocean Research](#)
- [Discover Artificial Intelligence](#)
- [International Journal of Transportation Science and Technology](#)
- [Journal of Geographical Systems](#)
- [Journal of Ocean Engineering and Science](#)
- [npj Sustainable Mobility and Transport](#)
- [Nuclear Engineering and Technology](#)
- [Progress in Disaster Science](#)
- [Scientific Reports](#)
- [Transportation Research Board](#)
- [Transportation Research Interdisciplinary Perspectives](#)
- [Transportation Research Part D: Transport and Environment](#)